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NUMERICAL CONTROL DEVICE AND DETERMINATION METHOD

Abstract

A numerical control device estimates the expansion and contraction amount of a ball screw. The numerical control device acquires a rotation angle and disturbance force of a Z-axis motor. The numerical control device acquires the torque of the Z-axis motor to acquire the disturbance force included in the torque. The numerical control device determines the mounted state of a tool on a main shaft on the basis of the expansion and contraction amount of the ball screw, and a peak angle decided by a change in a differential value, which is the time differential of the disturbance force.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application continuation of PCT International Application No. PCT/JP2023/041430, filed on Nov. 17, 2023, which claims priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2022-190645, filed on Nov. 29, 2022. Each of the above application(s) is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a numerical control device and a determination method.

Related Art

[0003] Japanese Patent Laid-Open No. 2022-56673 discloses a numerical control device in which it is determined whether a tool is mounted on a main shaft of a machine tool. The machine tool includes a main shaft, a main shaft head, and a Z-axis motor. The main shaft is rotatably provided on the main shaft head. The main shaft head is lifted and lowered by the drive of the Z-axis motor. During mounting of the tool on the main shaft, the main shaft head is lowered by the drive of the Z-axis motor. At this time, the numerical control device differentiates a torque applied to the Z-axis motor. Based on a differential value of the torque and a position of the main shaft head, the numerical control device decides a peak position of the differential value that varies according to the position of the main shaft head. The numerical control device performs statistical processing on the decided peak position, and determines a mounted state of the tool with respect to the main shaft based on a calculated statistical value.

[0004] In the machine tool, a feed shaft may be provided as a mechanism for lifting and lowering the main shaft head. The feed shaft extends in an up-down direction. The feed shaft is capable of positive and reverse rotation around an axis extending in the up-down direction by the drive of the Z-axis motor. A nut is fixed to the main shaft head. The nut is screwed onto the feed shaft. When the feed shaft rotates in the positive and reverse directions, the nut and the main shaft head to which the nut is fixed integrally rise and lower.

[0005] The feed shaft expands and contracts in the up-down direction due to a temperature change thereof. When the feed shaft expands and contracts, the peak position of the differential value of the torque calculated in order to determine the mounted state of the tool with respect to the main shaft varies. Thus, it cannot be specified whether the variation in the peak position is due to the influence of expansion and contraction of the feed shaft or the influence of the mounted state of the tool, and there is a risk that the numerical control device may not be able to accurately determine the mounted state of the tool.

SUMMARY

[0006] A numerical control device of one embodiment is a numerical control device that controls an operation of a machine tool, the machine tool including a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft and allow the tool gripped by a tool changer to be mounted on the main shaft. The numerical control device is characterized by including: an estimation part, estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition part, acquiring a torque of the motor in a mounting operation of the tool; and a determination part,

determining a mounted state of the tool with respect to the main shaft based on the expansion and contraction amount estimated by the estimation part and the torque acquired by the first acquisition part.

[0007] A determination method of another embodiment is a determination method for a mounted state of a tool with respect to a main shaft in a machine tool, the machine tool including a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft to allow the tool gripped by a tool changer to be mounted on the main shaft. The determination method is characterized by including: an estimation process for estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition process for acquiring a torque of the motor in a mounting operation of the tool; and a determination process for determining the mounted state based on the expansion and contraction amount estimated by the estimation process and the torque acquired by the first acquisition process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a machine tool 1.

[0009] FIG. 2 is a longitudinal sectional view of surroundings of a main shaft head 7.

[0010] FIG. 3 is a longitudinal sectional view of the interior of a main shaft 9.

[0011] FIG. 4 is a block diagram showing an electrical configuration of the machine tool 1 and a numerical control device 30.

[0012] FIG. 5 is a graph showing a disturbance force F applied to a Z-axis motor 51 and a differential value f , which vary according to a rotation angle θ of the Z-axis motor 51.

[0013] FIG. 6 is a graph showing thermal resistance R that varies according to a feed speed $V_{sub.m}$.

[0014] FIG. 7 is a conceptual diagram of a condition table 59.

[0015] FIG. 8 is a flowchart of parameter estimation processing.

[0016] FIG. 9 is a flowchart of second drive condition processing.

[0017] FIG. 10 is a flowchart showing the continuation of FIG. 9.

[0018] FIG. 11 is a graph showing a temporal change in an expansion and contraction amount $L_{sub.b}$ of a ball screw 41 specified in the parameter estimation processing and an estimation curve of the expansion and contraction amount $L_{sub.b}$.

[0019] FIG. 12 is a flowchart of period acquisition processing.

[0020] FIG. 13 is a flowchart of expansion and contraction amount estimation processing.

[0021] FIG. 14 is a flowchart of tool change processing.

[0022] FIG. 15 is a flowchart showing the continuation of FIG. 14.

DESCRIPTION OF THE EMBODIMENTS

[0023] The present disclosure provides a numerical control device and a determination method in which a mounted state of a tool with respect to a main shaft can be accurately determined.

[0024] A numerical control device of one embodiment is a numerical control device that controls an operation of a machine tool, the machine tool including a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft and allow the tool gripped by a tool changer to be mounted on the main shaft. The numerical control device is characterized by including: an estimation part, estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition part, acquiring a torque of the motor in a mounting operation of the tool; and a determination part,

determining a mounted state of the tool with respect to the main shaft based on the expansion and contraction amount estimated by the estimation part and the torque acquired by the first acquisition part.

[0025] The numerical control device estimates the expansion and contraction amount of the feed shaft according to a configuration for controlling the torque of the motor. Based on the estimated expansion and contraction amount and the torque of the motor, the numerical control device determines the mounted state of the tool with respect to the main shaft. Since the numerical control device takes into consideration the expansion and contraction amount, the mounted state of the tool with respect to the main shaft can be accurately determined.

[0026] In the numerical control device of another embodiment, the estimation part may include: a second acquisition part, acquiring information about the motor that drives when moving the main shaft in the predetermined direction via the feed shaft; and a calculation part, calculating the amount of heat entering and exiting the feed shaft based on the information acquired by the second acquisition part. The estimation part may estimate the expansion and contraction amount based on the amount of heat calculated by the calculation part. The numerical control device acquires the information about the motor that rotates the feed shaft, and estimates the expansion and contraction amount of the feed shaft based on the amount of heat calculated based on the acquired information. Accordingly, the numerical control device is able to accurately estimate the expansion and contraction amount of the feed shaft.

[0027] In the numerical control device of another embodiment, the second acquisition part may acquire, as the information, the torque and a rotation speed of the motor. The numerical control device acquires the torque and the rotation speed of the motor, and calculates the amount of heat entering and exiting the feed shaft based on the acquired torque and rotation speed. Accordingly, the numerical control device is able to accurately calculate the amount of heat entering and exiting the feed shaft.

[0028] In the numerical control device of another embodiment, the first acquisition part may further acquire the rotation angle of the motor. The determination part may include: a differentiation part, differentiating the torque acquired by the first acquisition part and calculating a differential value of the torque corresponding to the rotation angle; a first decision part, deciding, as a peak angle, the rotation angle corresponding to the differential value indicating either the maximum or the minimum among the differential values calculated by the differentiation part; a second decision part, deciding a corrected peak angle by correcting the peak angle decided by the first decision part based on the expansion and contraction amount estimated by the estimation part; and a correction determination part, determining the mounted state based on the corrected peak angle decided by the second decision part. The numerical control device corrects the peak angle based on the expansion and contraction amount of the feed shaft, and determines the mounted state of the tool with respect to the main shaft based on the corrected peak angle that has been corrected. Accordingly, since a deviation between the peak angle due to expansion and contraction of the feed shaft and an actual position of the main shaft can be suppressed, the numerical control device is able to accurately determine the mounted state of the tool with respect to the main shaft.

[0029] In the numerical control device of another embodiment, the determination part may include: a storage part, storing the corrected peak angle decided by the second decision part and the number of uses or a use time of the tool with respect to the main shaft in association with each other; a statistical processing part, performing statistical processing based on a plurality of the corrected peak angles stored by the storage part and calculating a statistical value; and a statistical determination part, determining the mounted state based on the statistical value calculated by the statistical processing part. The numerical control device determines the mounted state of the tool based on the statistical value based on multiple corrected peak angles. Accordingly, the numerical control device is able to eliminate the influence of variations in determining the corrected peak angle and accurately determine the mounted state.

[0030] In the numerical control device of another embodiment, in the case where a difference between the expansion and contraction amount estimated by the estimation part for the n -th (n is an integer equal to or greater than 2) time and the expansion and contraction amount estimated by the estimation part for the $(n-1)$ -th time is less than or equal to a threshold, the determination part may determine the mounted state; in the case where the difference is greater than the threshold, the determination part may not determine the mounted state. In the case where a change amount in the expansion and contraction amount of the feed shaft is excessively large, there is a risk that the estimation part may not be able to accurately estimate the expansion and contraction amount. In the numerical control device, by determining the estimated expansion and contraction amount, erroneous determination of the mounted state due to inaccurate estimation of the expansion and contraction amount can be suppressed.

[0031] In the numerical control device of another embodiment, the estimation part may estimate the expansion and contraction amount in an estimation period longer than a control period of the numerical control device. Generally, since the time required for expansion and contraction of the feed shaft is sufficiently long relative to the control period of the numerical control device, the numerical control device is able to accurately estimate the expansion and contraction amount of the feed shaft with a small amount of calculation.

[0032] In the numerical control device of another embodiment, the first acquisition part may acquire the torque and the rotation angle for each control period of the numerical control device. Since the torque and the rotation angle of the motor take positive and negative values, there is a risk that the positive and negative values may be averaged if an acquisition period is long and that a feature value may be lost. Thus, in the numerical control device, the torque and the rotation angle of the motor are acquired for each control period being a smallest unit at which acquisition is possible, and the mounted state of the tool with respect to the main shaft is determined based on the acquired torque and rotation angle. Accordingly, the numerical control device is able to accurately determine the mounted state of the tool with respect to the main shaft.

[0033] A determination method of another embodiment is a determination method for a mounted state of a tool with respect to a main shaft in a machine tool, the machine tool including a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft to allow the tool gripped by a tool changer to be mounted on the main shaft. The determination method is characterized by including: an estimation process for estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition process for acquiring a torque of the motor in a mounting operation of the tool; and a determination process for determining the mounted state based on the expansion and contraction amount estimated by the estimation process and the torque acquired by the first acquisition process. The determination method achieves similar effects to the numerical control device of the above embodiment.

[0034] An embodiment of the present disclosure will be described with reference to the drawings. The following description uses left and right, front and rear, and up and down indicated by arrows in the drawings. The left-right direction, the front-rear direction, and the up-down direction of a machine tool **1** are an X-axis direction, a Y-axis direction, and a Z-axis direction of the machine tool **1**, respectively. The right direction, forward direction, and upward direction are each a positive direction, and the left direction, rearward direction, and downward direction are each a negative direction. The machine tool **1** shown in FIG. **1** is a machine that performs cutting on a workpiece (not shown) with a tool **3**. A numerical control device **30** controls an operation of the machine tool **1**. The term “ATC” as mentioned in the present embodiment is an abbreviation for “automatic tool changer”, and the term “NC” as mentioned in the present embodiment is an abbreviation for “numerical control”.

Structure of Machine Tool **1**

As shown in FIG. 1 and FIG. 2, the machine tool **1** includes a base **2**, a column **5**, a control box **6**, a table device **10**, a main shaft head **7**, a main shaft **9**, and a tool changer **20**. The base **2** is a metal foundation of a substantially rectangular parallelepiped shape. The column **5** is fixed to a rear part of an upper surface of the base **2**. The column **5** is a pillar extending in the up-down direction. The control box **6** is fixed to a back surface of the column **5**. The control box **6** houses the numerical control device **30**.

[0035] The table device **10** includes a Y-axis movement mechanism (not shown), a Y-axis table **12**, a table **13**, and an X-axis movement mechanism (not shown). The Y-axis movement mechanism is provided on the upper surface of the base **2** and in front of the column **5**, and includes a Y-axis motor **54** (refer to FIG. 4). The Y-axis movement mechanism moves the Y-axis table **12** in the Y-axis direction in response to the drive of the Y-axis motor **54**. The X-axis movement mechanism is provided on an upper part of the Y-axis table **12** and includes an X-axis motor **53** (refer to FIG. 4). The X-axis movement mechanism moves the table **13** in the X-axis direction in response to the drive of the X-axis motor **53**. Hence, the table **13** is movable in the X-axis direction and Y-axis direction above the base **2** by means of the X-axis movement mechanism and the Y-axis movement mechanism.

Structure of Z-Axis Movement Mechanism **8**

As shown in FIG. 2, a Z-axis movement mechanism **8** is provided on a front part of the column **5**. The Z-axis movement mechanism **8** includes a ball screw **41**, bearings **42** and **43**, a cam follower **49**, and a Z-axis motor **51** (refer to FIG. 4). The ball screw **41** is provided on the front part of the column **5** and extends in the Z-axis direction. The bearings **42** and **43** are inserted through the ball screw **41** and rotatably support the ball screw **41**. The bearing **42** is provided above the bearing **43**. The cam follower **49** is provided in front of the bearing **42**. The cam follower **49** slides on a cam surface of a plate cam **47** of the main shaft head **7** described later.

[0036] The Z-axis motor **51** is fixed above the bearing **42**. The ball screw **41** is connected to an output shaft of the Z-axis motor **51** through a coupling (not shown). The ball screw **41** is rotated positively and reversely around an axis extending in the up-down direction by the drive of the Z-axis motor **51**. A nut **44** is screwed onto the ball screw **41** between the bearing **42** and the bearing **43**. The nut **44** is fixed to a rear end of the main shaft head **7**. When the Z-axis motor **51** rotates in the positive direction, the ball screw **41** rotates, and the nut **44** and the main shaft head **7** integrally rise. When the Z-axis motor **51** rotates in the reverse direction, the ball screw **41** rotates reversely, and the nut **44** and the main shaft head **7** integrally lower.

Internal Structure of Main Shaft Head **7**

As shown in FIG. 2, the main shaft head **7** is provided on the front part of the column **5**. The main shaft head **7** is of a box shape. The main shaft head **7** includes a main shaft motor **52**, a crank lever **45**, a support shaft **46**, and a coil spring **48**. The main shaft motor **52** is fixed to a front part of an upper surface of the main shaft head **7**.

[0037] The support shaft **46** is fixed to the inside of a rear part of the main shaft head **7**. The support shaft **46** is of a rod shape extending in the left-right direction. The crank lever **45** is provided on an upper part of the main shaft head **7**. The crank lever **45** is a plate of a substantially L-shape when viewed from the left side. The support shaft **46** is inserted through a bent portion of the crank lever **45**. The crank lever **45** is swingable about the support shaft **46**. The crank lever **45** includes the plate cam **47**. The cam surface is formed on a back surface of the plate cam **47**. The cam surface of the plate cam **47** is able to come into and out of contact with the cam follower **49**. The coil spring **48** is provided inside a rear part of the main shaft head **7**. The coil spring **48** extends in the front-rear direction. One end of the coil spring **48** is fixed to a back surface of the main shaft head **7**. A rear end of the coil spring **48** is fixed to a rear end of the crank lever **45** below the plate cam **47**. The coil spring **48** constantly biases the crank lever **45** clockwise when viewed from the right side.

Internal Structure of Main Shaft **9**

As shown in FIG. 2 and FIG. 3, the main shaft 9 is provided inside a front lower part of the main shaft head 7 and is of a cylindrical shape extending in the up-down direction. The main shaft 9 is supported by the main shaft head 7. The main shaft 9 is rotatably supported around an axis extending in the up-down direction. The main shaft 9 is connected to an output shaft of the main shaft motor 52. The main shaft 9 is rotated positively and reversely around an axis extending in the up-down direction by the drive of the main shaft motor 52.

[0038] The main shaft 9 includes a shaft hole 91, a mounting hole 92, a space 93, a sliding hole 94, a clamp shaft 81, and a spring 82. The shaft hole 91 is formed across from an upper end to a lower part of the main shaft 9 and passes through the center of the main shaft 9. The mounting hole 92 is formed at a lower end of the main shaft 9. The tool 3 is mounted in the mounting hole 92. The space 93 is formed in communication with an upper part of the mounting hole 92. The sliding hole 94 is formed to communicate between the shaft hole 91 and the space 93.

[0039] The clamp shaft 81 is provided inside the shaft hole 91 and is movable in the up-down direction with respect to the shaft hole 91. The clamp shaft 81 includes a support 83, a shaft 84, and a gripper 85. The support 83 is provided at an upper end of the clamp shaft 81 and is of a columnar shape. The support 83 supports a pin 95. The pin 95 is provided protruding from the clamp shaft 81. The pin 95 is able to come into and out of contact with a front end of the crank lever 45 located above the pin 95.

[0040] The shaft 84 is of a columnar shape extending downward from the support 83. The gripper 85 is provided at a lower end of the shaft 84 and includes multiple steel balls (not shown). The spring 82 is provided inserted inside the shaft hole 91. An upper end of the spring 82 engages with the support 83. The spring 82 constantly biases the clamp shaft 81 upward.

[0041] The main shaft 9 is capable of mounting the tool 3. The tool 3 includes a holder 14 and a cutting tool 4. One end of the holder 14 holds the cutting tool 4, and the other end of the holder 14 includes a mounting part 15 and a pull stud 17. The mounting part 15 is of a conical shape reducing in diameter toward the other end of the holder 14. The pull stud 17 protrudes from a top part of the mounting part 15 toward the other end of the holder 14. The mounting part 15 is mounted in the mounting hole 92.

[0042] When the clamp shaft 81 moves downward against elastic force of the spring 82, the gripper 85 exits from the sliding hole 94 into the space 93. The gripper 85 releases its grip on the pull stud 17 of the holder 14. When the clamp shaft 81 moves upward due to the elastic force of the spring 82 from the state of having moved down, the gripper 85 moves from the space 93 into the sliding hole 94. The gripper 85 grips the pull stud 17 with multiple steel balls.

Structure of Tool Changer 20

As shown in FIG. 2, the tool changer 20 is provided in front of the main shaft head 7. The tool changer 20 is of a turret type. The tool changer 20 includes a tool magazine 21, multiple grip arms 23, and a magazine motor 55. The tool magazine 21 is of a substantially disk shape. The tool magazine 21 is rotatably supported about a support shaft 25 extending forward and downward. Each of the grip arms 23 is provided at predetermined intervals on an outer circumference of the tool magazine 21. The multiple grip arms 23 are swingably provided in the front-rear direction of the tool magazine 21. A tip of the grip arm 23 detachably grips the holder 14 of the tool 3. An output shaft of the magazine motor 55 is connected to the support shaft 25. The tool magazine 21 is rotated positively and reversely about the support shaft 25 by the drive of the magazine motor 55.

Operations of Mounting, Detaching, and Replacing Tool 3

As shown in FIG. 2, with the mounting part 15 of the holder 14 mounted in the mounting hole 92, based on a control command from the numerical control device 30, the Z-axis motor 51 rotates in the positive direction. The ball screw 41 rotates around its axis, and the main shaft head 7 and the main shaft 9 rise from a machining origin of the table 13. The machining origin is a mechanical origin and is an uppermost position where the workpiece can be machined.

[0043] When the main shaft head 7 and the main shaft 9 rise, the cam follower 49 contacts the plate

cam **47** of the crank lever **45** and slides along the cam surface of the plate cam **47**. The crank lever **45** rotates counterclockwise about the support shaft **46** when viewed from the right side, against the elastic force of the coil spring **48**. The front end of the crank lever **45** engages with the pin **95** from above and presses the clamp shaft **81** downward. The clamp shaft **81** biases the gripper **85** downward against the elastic force of the spring **82**. The gripper **85** releases its grip on the pull stud **17**.

[0044] The main shaft **9** further rises to an ATC origin. The ATC origin is a position of the main shaft **9** and is located above the machining origin. In the case where the main shaft **9** is located at the ATC origin, the tool magazine **21** is rotatable.

[0045] When the main shaft **9** rises, the tool **3** detaches from the mounting hole **92** of the main shaft **9**. The tool **3** that has detached from the main shaft **9** is referred to as first tool. Among the multiple grip arms **23** of the tool changer **20**, one grip arm **23** (hereinafter referred to as first grip arm) at a tool change position grips the first tool that has detached from the main shaft **9**. The tool change position is a lowermost position of the tool magazine **21** and is a position facing and close to the main shaft **9**.

[0046] When the main shaft head **7** reaches the ATC origin, based on the control command from the numerical control device **30**, the tool changer **20** rotates the tool magazine **21** by the drive of the magazine motor **55**. The tool changer **20** indexes the tool **3** (hereinafter referred to as second tool) that holds the cutting tool **4** designated by a control command of an NC program to the tool change position. At this time, the tool magazine **21** rotates from a state in which the first tool is at the tool change position to a state in which the second tool is at the tool change position. The tool magazine **21** rotates in a state in which the first tool is gripped by the first grip arm and the second tool is gripped by another grip arm **23** (hereinafter referred to as second grip arm). The second tool indexed to the tool change position is located below the main shaft head **7** that has moved to the ATC origin.

[0047] Next, based on the control command from the numerical control device **30**, the Z-axis motor **51** rotates in the reverse direction. The ball screw **41** rotates reversely around its axis, and the main shaft **9** is lowered from the ATC origin. The mounting part **15** of the second tool enters the mounting hole **92** of the main shaft **9**.

[0048] With the mounting part **15** inserted into the mounting hole **92**, the main shaft **9** and the main shaft head **7** further lower. The cam follower **49** slides along the cam surface of the plate cam **47** and then separates from the plate cam **47**. Due to the elastic force of the coil spring **48**, the crank lever **45** rotates clockwise about the support shaft **46** when viewed from the right side. The front end of the crank lever **45** separates from the pin **95**, releasing the downward pressing on the clamp shaft **81**. The clamp shaft **81** releases the downward bias on the gripper **85**. The gripper **85** moves from the space **93** to the sliding hole **94**. The gripper **85** grips and pulls up the pull stud **17** of the second tool with multiple steel balls. The second grip arm of the tool changer **20** releases its grip on the holder **14** of the second tool. The mounting of the mounting part **15** into the mounting hole **92** of the main shaft **9** is completed, and the mounting of the second tool to the main shaft **9** is completed. An operation of the clamp shaft **81** and the crank lever **45** when the tool **3** is mounted to the main shaft **9** is referred to as a mounting operation. An operation of the clamp shaft **81** and the crank lever **45** when the tool **3** is detached from the main shaft **9** is referred to as a detachment operation. The main shaft head **7**, the crank lever **45** or the like that are driven by rotation of the Z-axis motor **51** in the mounting operation are referred to as a mounting mechanism.

Electrical Configuration of Numerical Control Device **30** and Machine Tool **1**

As shown in FIG. **4**, the numerical control device **30** includes a CPU **31**, a ROM **32**, a RAM **33**, a storage device **34**, an input/output interface **35**, and drive circuits **61** to **65**. The CPU **31** performs integrated control of the numerical control device **30**. The ROM **32** stores various programs. The RAM **33** stores various data during execution of processing.

[0049] The storage device **34** is a non-volatile memory and stores the NC program, a condition

table 59 described later, and the like. The NC program consists of multiple blocks. Each block includes at least one command such as a tool change command. For each tool 3, the storage device 34 stores the number of times of mounting (equivalent to the number of uses) the tool 3 on the main shaft 9. In the case where the mounting operation of the tool 3 with respect to the main shaft 9 is performed, the CPU 31 adds 1 to the number of times of mounting corresponding to the tool 3 (second tool) on which the mounting operation is performed. In the case where the tool 3 mounted on the main shaft 9 is detached and is replaced with a different tool 3, the CPU 31 sets the number of times of mounting corresponding to the detached tool 3 to 0.

[0050] The input/output interface 35 is electrically connected to the drive circuits 61 to 65, encoders 71 to 75, an input part 37, and a display part 38, and performs input and output of various signals. The input part 37 and the display part 38 are provided on an operation panel 36 of the machine tool 1. The operation panel 36 is provided on an outer wall of a cover (not shown) that covers the machine tool 1. The input part 37 accepts input of various information, operation instructions or the like, and inputs them to the CPU 31 via the input/output interface 35. Based on commands input from the CPU 31 via the input/output interface 35, the display part 38 displays various screens, abnormality information, and the like.

[0051] The drive circuit 61 is electrically connected to the Z-axis motor 51. The drive circuit 62 is electrically connected to the main shaft motor 52. The drive circuit 63 is electrically connected to the X-axis motor 53. The drive circuit 64 is electrically connected to the Y-axis motor 54. The drive circuit 65 is electrically connected to the magazine motor 55. Based on commands input from the CPU 31 via the input/output interface 35, the drive circuits 61, 62, 63, 64, and 65 output drive currents to the Z-axis motor 51, the main shaft motor 52, the X-axis motor 53, the Y-axis motor 54, and the magazine motor 55, respectively. The Z-axis motor 51, the main shaft motor 52, the X-axis motor 53, the Y-axis motor 54, and the magazine motor 55 are servo motors that rotate according to the input drive currents. The drive circuits 61, 62, 63, 64, and 65 output the drive currents as return values to the input/output interface 35.

[0052] The Z-axis motor 51 includes an encoder 71. The main shaft motor 52 includes an encoder 72. The X-axis motor 53 includes an encoder 73. The Y-axis motor 54 includes an encoder 74. The magazine motor 55 includes an encoder 75. The encoders 71, 72, 73, 74, and 75 are absolute value encoders, and detect rotation angles of the Z-axis motor 51, the main shaft motor 52, the X-axis motor 53, the Y-axis motor 54, and the magazine motor 55, respectively. The encoders 71, 72, 73, 74, and 75 input the detected rotation angles to the input/output interface 35. By acquiring the rotation angle of the Z-axis motor 51 detected by the encoder 71, the CPU 31 estimates a rotation angle of the ball screw 41 connected to the Z-axis motor 51, and estimates a position of the main shaft 9 in the Z-axis direction that moves due to rotation of the ball screw 41.

Disturbance Force in Mounting Operation and Variation in Differential Value of Disturbance Force (A) of FIG. 5 is a graph showing a relationship between the rotation angle of the Z-axis motor 51 (horizontal axis) and disturbance force applied to the Z-axis motor 51 (vertical axis) in the mounting operation. In (A) of FIG. 5, to eliminate the influence of vibration of the machine tool 1 or the like, low-pass filter processing that removes natural vibration of the machine tool 1 is applied. In the mounting operation, the rotation angle of the Z-axis motor 51 changes in the negative direction.

[0053] The disturbance force applied to the Z-axis motor 51 is a force acting on the Z-axis motor 51 as a reaction force to the drive of the Z-axis motor 51. The disturbance force is included in a torque of the Z-axis motor 51 and satisfies a relationship in the following Expression 1.

$$[00001] \quad T_m = F + J^{\text{Math.}} + D \dot{\theta} + k \theta \quad [\text{Expression1}]$$

In Expression 1, T.sub.m represents the torque (Nm) of the Z-axis motor 51. F represents the disturbance force (Nm) applied to the Z-axis motor 51. θ represents the rotation angle (rad) of the Z-axis motor 51. “ $\dot{\theta}$ (with one dot above)” represents a first-order time differential (angular speed

(rad/s)) of the rotation angle θ . " $\ddot{\theta}$ (with two dots above)" represents a second-order time differential (angular acceleration (rad/s²)) of the rotation angle θ . J (kg.Math.m²) represents a moment of inertia related to the Z-axis motor **51**. The second term on the right side represents inertial force. D represents a viscosity coefficient (Nm/(rad/s)) related to the Z-axis motor **51**. The third term on the right side represents viscous force. k represents a coefficient (Nm) related to Coulomb friction related to the Z-axis motor **51**. The fourth term on the right side represents Coulomb friction force. The magnitudes of the moment of inertia J , viscosity coefficient D , and coefficient k depend on a rotor portion (not shown) inside the Z-axis motor **51** that rotates with the operation of the Z-axis motor **51**, the ball screw **41**, the nut **44** that moves up and down with the operation of the Z-axis motor **51**, and the main shaft head **7** or the like. The details thereof are omitted.

[0054] The CPU **31** acquires the drive current of the Z-axis motor **51** from the drive circuit **61** and acquires the rotation angle θ of the Z-axis motor **51** from the encoder **71**, thereby acquiring the torque $T_{sub.m}$ of the Z-axis motor **51**. The CPU **31** acquires the disturbance force F based on the torque $T_{sub.m}$ and rotation angle θ of the Z-axis motor **51**.

[0055] The CPU **31** acquires the rotation angle θ of the Z-axis motor **51** from the encoder **71**. The CPU **31** estimates the position of the main shaft **9** in the Z-axis direction based on the acquired rotation angle θ and the lead of the ball screw **41**.

[0056] (B) of FIG. 5 is a graph showing a relationship between the rotation angle θ of the Z-axis motor **51** (horizontal axis) and a time differential value (hereinafter referred to as differential value f) of the disturbance force F (vertical axis) in the mounting operation. In (B) of FIG. 5, the rotation angle θ of the Z-axis motor **51** in the case where the differential value f becomes minimum in the mounting operation is referred to as peak angle ϕ . A timing at which the differential value f becomes minimum (that is, rotation angle θ becomes peak angle ϕ) in the mounting operation is a timing at which the plate cam **47** separates from the cam follower **49**.

Mounted State of Tool **3** With Respect to Main Shaft **9**

A state in which the tool **3** is mounted on the main shaft **9** in a manner that allows machining of the workpiece with the tool **3** and does not affect cutting accuracy is referred to as a completely mounted state. In contrast, in the case where the machine tool **1** performs cutting in a state (hereinafter referred to as incompletely mounted state) in which the tool **3** is incompletely mounted on the main shaft **9**, there is a possibility that the cutting accuracy may decrease. The incompletely mounted state is a state in which the tool **3** is mounted on the main shaft **9** while foreign matter (for example, chips generated in cutting) remains adhering between the mounting part **15** and the mounting hole **92**. In the incompletely mounted state, a position where the gripper **85** pulls up and grips the pull stud **17** in the mounting operation is lower than that in the completely mounted state. The completely mounted state and the incompletely mounted state are collectively referred to as the mounted state.

[0057] The position where the gripper **85** grips the pull stud **17** changes depending on the mounted state of the tool **3** on the main shaft **9**. Accordingly, a timing at which the plate cam **47** contacts the cam follower **49** in the mounting operation or a timing at which the plate cam **47** detaches from the cam follower **49** in the detachment operation changes according to the mounted state of the tool **3** with respect to the main shaft **9**. Hence, the peak angle ϕ changes according to the mounted state. The numerical control device **30** determines the mounted state based on a change in the peak angle ϕ for each mounting operation.

Determination of Mounted State

In order to determine the mounted state, the CPU **31** of the numerical control device **30** acquires the rotation angle θ of the Z-axis motor **51** and the disturbance force F applied to the Z-axis motor **51** in the mounting operation every period $t_{sub.m}$ (s). In the present embodiment, the period $t_{sub.m}$ is $t_{sub.m}=0.5$ ms, which is a shortest time among control periods in the numerical control device **30**.

[0058] The CPU **31** applies a low-pass filter to the acquired disturbance force F , and then performs time differentiation and calculates the differential value f . The CPU **31** stores the calculated differential value f and the rotation angle θ in association with each other in the RAM **33**. The CPU **31** decides the peak angle ϕ from a variation (refer to (B) of FIG. 5) in the differential value f stored in the RAM **33**. Every time the mounting operation is executed, the CPU **31** associates the decided peak angle ϕ with the number of times of mounting the tool **3**, and sequentially stores them in the storage device **34**.

[0059] Every time the peak angle ϕ is stored in the storage device **34**, the CPU **31** performs statistical processing on the stored peak angle and calculates a statistical value. More specifically, the CPU **31** decides a target peak angle from the peak angles ϕ stored in the storage device **34**. The target peak angle is a peak angle excluding a latest peak angle (hereinafter referred to as first peak angle) and the peak angle ϕ in the case where the incompletely mounted state is reached. The CPU **31** extracts a predetermined number of peak angles ϕ counting from a latest target peak angle (hereinafter referred to as second peak angle) among the target peak angles. The CPU **31** calculates, as statistical values, an average value μ and a standard deviation σ of the extracted predetermined number of peak angles ϕ . If the first peak angle (that is, the peak angle ϕ in a latest mounting operation) is greater than the sum ($\mu + 6\sigma$) (hereinafter referred to as determination threshold $\phi_{\text{sub.max}}$) of the average value μ and 6 times the standard deviation σ , the CPU **31** determines that the mounted state after the mounting operation is an incompletely mounted state.

Expansion and Contraction of Ball Screw **41**

The ball screw **41** of the Z-axis movement mechanism **8** that moves the main shaft **9** in the Z-axis direction is heated by heat generated from the Z-axis motor **51** and frictional heat with the nut **44**. The ball screw **41** dissipates heat due to a temperature difference with the outside air. Hence, the ball screw **41** changes temperature by being heated from the outside and dissipating heat to the outside. When a temperature change occurs, the ball screw **41** expands and contracts particularly in the Z-axis direction which is a longitudinal direction.

[0060] When the ball screw **41** expands and contracts, the timing at which the plate cam **47** separates from the cam follower **49** changes according to an expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw **41**. That is, the peak angle ϕ changes according to the expansion and contraction amount $L_{\text{sub.b}}$. In the determination of the mounted state based on the peak angle ϕ , it is difficult to distinguish between a variation in the peak angle ϕ due to the expansion and contraction of the ball screw **41** and a variation in the peak angle ϕ due to the incompletely mounted state. Hence, when determining the mounted state of the tool **3** with respect to the main shaft **9**, the CPU **31** estimates the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw **41** in the Z-axis direction and corrects the peak angle ϕ based on the expansion and contraction amount $L_{\text{sub.b}}$.

Estimation of Expansion and Contraction Amount $L_{\text{SUB.b}}$ of Ball Screw **41**

The CPU **31** estimates the expansion and contraction amount $L_{\text{sub.b}}$ (m) of the ball screw **41** in the Z-axis direction using a mathematical model. The CPU **31** calculates a temperature T_b (K) of the ball screw **41** in an estimation period $t_{\text{sub.s}}$ (s). In the present embodiment, $t_{\text{sub.s}} = 30$ s. The temperature $T_b(i)$ of the ball screw **41** in the i -th (i is an integer equal to or greater than 1) sampling which is latest is expressed by Expression 2 using the temperature $T_b(i-1)$ (K) of the ball screw **41** in the $(i-1)$ -th sampling and a temperature change ΔT_b (K) of the ball screw **41** from the $(i-1)$ -th to the i -th sampling.

$$[00002] \quad T_b(i) = T_b(i-1) + \Delta T_b \quad [\text{Expression 2}]$$

In the case where $i=1$, the temperature $T_b(i-1)$ of the ball screw **41** in the sampling, that is, the initial temperature $T_b(0)$ of the ball screw **41**, is assumed to be the same as a temperature T_a (K) of the outside air. For simplification of calculation, it is assumed that $T_a = 0$ (K).

[0061] The temperature change ΔT_b is expressed by the following Expression 3.

$$[00003] \quad T_b = (Q_m + Q_t - Q_{ba}) / H_b \quad [\text{Expression3}]$$

In Expression 3, $Q_{\text{sub.m}}$ represents the amount of heat (J) added to the ball screw **41** due to the heat generation of the Z-axis motor **51** in the estimation period $t_{\text{sub.s}}$. $Q_{\text{sub.f}}$ represents the amount of heat (J) added to the ball screw **41** due to friction in the estimation period $t_{\text{sub.s}}$. $Q_{\text{sub.ba}}$ represents the amount of heat (J) released to the outside air in the estimation period $t_{\text{sub.s}}$. H_b represents a heat capacity (J/K) of the ball screw **41**. The heat capacity H_b is calculated based on the specific heat of the ball screw **41** and the mass of the ball screw **41**.

[0062] The amount of heat $Q_{\text{sub.m}}$ is expressed by the following Expression 4.

$$[00004] \quad Q_m = Pr \times (L_m)^2 \times t_s \quad [\text{Expression4}]$$

In Expression 4, Pr represents a coefficient (J/Nm.sup.2) that expresses the amount of heat added to the ball screw **41** due to the heat generation of the Z-axis motor **51**. L_m represents an average load (Nm) applied to the Z-axis motor **51** in the estimation period $t_{\text{sub.s}}$, and is expressed by the following Expression 5.

$$[00005] \quad L_m = T / t_s \quad [\text{Expression5}]$$

In Expression 5, ΣT is an integration of an absolute value of the torque of the Z-axis motor **51** in the estimation period $t_{\text{sub.s}}$ (Nm.Math.s), and is expressed by the following Expression 6.

$$[00006] \quad \text{.Math. } T = \text{.Math.}_{0}^{t_s} (\text{.Math. } T_m \text{ .Math. } \times t_m) \quad [\text{Expression6}]$$

[0063] The amount of heat $Q_{\text{sub.f}}$ is expressed by the following Expression 7.

$$[00007] \quad Q_f = F_c \times TD \quad [\text{Expression7}]$$

In Expression 7, F_c represents a coefficient (J/(Nm.Math.m)) representing the amount of heat applied to the ball screw **41** due to friction. ETD is an integration of the product of the absolute value of the torque $T_{\text{sub.m}}$ applied to the Z-axis motor **51** and an absolute value of a movement distance of the main shaft **9** in the period $t_{\text{sub.m}}$ in the estimation period $t_{\text{sub.s}}$ (Nm.Math.m), and is expressed by the following Expression 8.

[00008]

$$\text{.Math. } TD = \text{.Math.}_{0}^{t_s} (\text{.Math. } T_m \text{ .Math. } \times \text{.Math. } \frac{V_m}{60 \times 1000} \times t_m \text{ .Math. }) \quad [\text{Expression8}]$$

In Expression 8, $V_{\text{sub.m}}$ is a feed speed (mm/min) of the main shaft **9** in the Z-axis direction. $V_{\text{sub.m}}$ is decided based on the rotation angle θ of the Z-axis motor **51** detected by the encoder **71** and the lead of the ball screw **41**.

[0064] The amount of heat $Q_{\text{sub.ba}}$ is expressed by the following Expression 9.

$$[00009] \quad Q_{ba} = (T_b - T_a)t_s / R_{ba} \quad [\text{Expression9}]$$

In Expression 9, the temperature T_a of the outside air is assumed to be 0 (K) as mentioned above. R_{ba} represents an average thermal resistance (sK/J) in the estimation period $t_{\text{sub.s}}$, and is expressed by the following Expression 10.

$$[00010] \quad R_{ba} = R / t_s \quad [\text{Expression10}]$$

In Expression 10, ΣR is an integration (s2K/J) of the thermal resistance R (sK/J) of the ball screw **41** calculated in the estimation period $t_{\text{sub.s}}$, and is expressed by the following Expression 11.

$$[00011] \quad \text{.Math. } R = \text{.Math.}_{0}^{t_s} (R \times t_m) \quad [\text{Expression11}]$$

[0065] FIG. 6 is a graph obtained by estimating the thermal resistance R from the ball screw **41** with respect to the air around based on experimental results. The thermal resistance R varies based on the feed speed $V_{\text{sub.m}}$ of the main shaft **9** in the Z-axis direction. More specifically, in the case where the feed speed $V_{\text{sub.m}}$ is less than a predetermined boundary speed V_{as} , the magnitude of the thermal resistance R is almost constant at $R_{\text{sub.max}}$. On the other hand, when the feed speed $V_{\text{sub.m}}$ becomes equal to or greater than the boundary speed $V_{\text{sub.ds}}$, the thermal resistance R decreases as the feed speed $V_{\text{sub.m}}$ increases. The reason is thought to be that the rotation speed of the ball screw **41** rotated by the drive of the Z-axis motor **51** increases, and turbulence occurs in the

air around the ball screw **41**. Generally, it is known that when turbulence occurs at an interface between a solid and a fluid, the thermal resistance significantly decreases. At a maximum feed speed $V_{sub.m}$ (hereinafter referred to as fast forward speed $V_{sub.max}$), the magnitude of the thermal resistance R becomes $R_{sub.min}$. In the present embodiment, the magnitude of the thermal resistance R is approximated as $R_{sub.max}$ in the case where the feed speed $V_{sub.m}$ is less than the boundary speed $V_{sub.ds}$. In the case where the feed speed $V_{sub.m}$ is equal to or greater than the boundary speed $V_{sub.ds}$, the thermal resistance R is approximated to change proportionally with a proportionality constant a .

[0066] As described above, the thermal resistance R is expressed by the following Expression 12.

$$[00012] \begin{cases} R = R_{max} & (V_m < V_{ds}) \\ R = aV_m + b & (V_m \geq V_{ds}) \end{cases} \quad [\text{Expression12}]$$

In Expression 12, the proportionality constant a is expressed by the following Expression 13, and an intercept b is expressed by the following Expression 14.

$$[00013] a = (R_{min} - R_{max}) / (V_{max} - V_{ds}) \quad [\text{Expression13}]$$

$$b = R_{max} - (R_{min} - R_{max})V_{ds} / (V_{max} - V_{ds}) \quad [\text{Expression14}]$$

[0067] The temperature $Tb(i-1)$ in the $(i-1)$ -th time is known, and if the temperature change ΔTb is calculated based on Expressions 3 to 14, the temperature $Tb(i)$ can be calculated according to Expression 2. The expansion and contraction amount $L_{sub.b}$ of the ball screw **41** in the case where the ball screw **41** changes from the initial temperature (assuming $Tb(0)=Ta=0$ (K)) to the temperature $Tb(i)$ is estimated according to the following Expression 15.

$$[00014] L_b = \alpha \times Tb(i) \quad [\text{Expression15}]$$

In Expression 15, α is an expansion coefficient ($\mu\text{m/K}$) of the ball screw **41**, and is calculated as the product of a linear expansion coefficient ($1/\text{K}$) of the ball screw **41** and an effective length (m) of the ball screw **41**. The effective length of the ball screw **41** is a distance in the Z-axis direction between the bearing **43** (refer to FIG. 2) and the ATC origin.

Estimation of Parameter Necessary for Estimating Expansion and Contraction Amount $L_{SUB.b}$

To estimate the expansion and contraction amount $L_{sub.b}$ of the ball screw **41** according to Expression 15, the numerical control device **30** estimates a parameter necessary for estimating the expansion and contraction amount $L_{sub.b}$. Here, the parameters to be estimated include the coefficients Pr and Fc , the thermal resistances $R_{sub.max}$ and $R_{sub.min}$, and the boundary speed $V_{sub.ds}$.

[0068] To estimate the parameters, the numerical control device **30** specifies the expansion and contraction amount $L_{sub.b}$ by performing a tool change multiple times. More in detail, by acquiring the disturbance force F (refer to (A) of FIG. 5) that varies according to the rotation angle θ for each execution of the mounting operation in the tool change, the numerical control device **30** calculates the differential value f (refer to (B) of FIG. 5) and decides the peak angle ϕ .

[0069] The numerical control device **30** calculates a difference p between the peak angle ϕ decided in a latest tool change and the peak angle ϕ (hereinafter referred to as reference peak angle $\phi_{sub.b}$) as a reference. In the present embodiment, the reference peak angle $\phi_{sub.b}$ is the peak angle ϕ decided in the first tool change in the parameter estimation.

[0070] From the peak angle, the position of the main shaft **9** at the timing at which the plate cam **47** separates from the cam follower **49** is estimated. $\phi_{sub.b}$ indicates the position of the main shaft **9** at the timing at which the plate cam **47** separates from the cam follower **49** in the mounting operation performed for the first time. ϕ indicates the position of the main shaft **9** at the timing at which the plate cam **47** separates from the cam follower **49** in the latest mounting operation. That is, the difference p indicates a change amount from the position of the main shaft **9** at the first time to the position of the main shaft **9** at the latest time at the same timing. The numerical control device **30** calculates a change amount in the position of the main shaft **9** in the Z-axis direction

based on the difference $p (= \varphi - \varphi_{\text{sub.b}})$ of the peak angle φ , and specifies this change amount as the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw **41** in the Z-axis direction.

[0071] By this specification method, the expansion and contraction amount $L_{\text{sub.b}}$ in a tool change can be specified. This specification method is on the premise that the tool change is normally performed. For example, in workpiece machining, there is a possibility that the tool change may not be normally performed due to effects such as chips becoming caught between the main shaft **9** and the tool **3**. Thus, the expansion and contraction amount $L_{\text{sub.b}}$ cannot be specified by this specification method. Hence, it is necessary to estimate the expansion and contraction amount $L_{\text{sub.b}}$ using Expression 15.

[0072] During parameter estimation, the numerical control device **30** performs control based on the condition table **59** shown in FIG. 7. The condition table **59** includes seven blocks. The condition table **59** defines a control command for each block. The numerical control device **30** executes the control commands in order from block No. 1. That is, the block No. indicates the processing order of each block. The control commands include a first reciprocation command, a second reciprocation command, a third reciprocation command, a stop command, an end command, and so on.

[0073] The control command is associated with a first drive condition of the Z-axis motor **51**, a second drive condition of the Z-axis motor **51**, and a time during which the control command is continuously executed. In each control command, the numerical control device **30** drives the Z-axis motor **51** under the first drive condition for a predetermined time (for example, 5 minutes), and when the predetermined time elapses, drives the Z-axis motor **51** under the second drive condition, and again drives the Z-axis motor **51** under the first drive condition for the predetermined time. The numerical control device **30** drives the Z-axis motor **51** under the first drive condition and the second drive condition for a time (4 hours) indicated by the control command.

[0074] In the first reciprocation command, based on the first drive condition, the numerical control device **30** drives the Z-axis motor **51** to reciprocate the main shaft **9** at a feed speed $F5000$ ($V_{\text{sub.m}} = 5000 \text{ mm/min}$) in the Z-axis direction within a machining area for the predetermined time. The machining area is an area lower than the machining origin and is an area where the main shaft **9** moves during workpiece machining by the machine tool **1**. The descriptions of the second reciprocation command and the third reciprocation command are omitted. In the stop command, based on the first drive condition, the numerical control device **30** stops the Z-axis motor **51** to stop the main shaft **9** at the machining origin for the predetermined time.

[0075] In the first reciprocation command, second reciprocation command, third reciprocation command, and stop command (block Nos. 1 to 6), the numerical control device **30** drives the Z-axis motor **51** to execute a tool change as the second drive condition. The numerical control device **30** decides the peak angle φ and specifies the expansion and contraction amount $L_{\text{sub.b}}$ in the mounting operation of the tool **3**. That is, in block Nos. 1 to 6, the expansion and contraction amount $L_{\text{sub.b}}$ is specified every predetermined time. The end command (block No. 7) is a command for ending the parameter estimation.

Parameter Estimation Processing

Referring to FIG. 8, the parameter estimation processing executed by the CPU **31** of the numerical control device **30** is described. In the parameter estimation processing, the CPU **31** estimates a parameter necessary for estimating the expansion and contraction amount $L_{\text{sub.b}}$.

[0076] An operator inputs a parameter estimation command to the input part **37**. The input part **37** accepts the input parameter estimation command and inputs it to the CPU **31**. In the case where the parameter estimation command is input, the CPU **31** reads out and executes a control program stored in the ROM **32**, thereby starting the parameter estimation processing.

[0077] At the start of the parameter estimation processing, the main shaft **9** is located at the machining origin. A variable K that may be used in the parameter estimation processing is an integer equal to or greater than 1, and indicates a block No. in the condition table **59** (refer to FIG.

7). The variable K is stored in the RAM 33 and has a value of 1 when the parameter estimation processing is started.

[0078] The CPU 31 reads one block from the condition table 59 (S1). The CPU 31 determines whether a control command of the read block is an end command (S2). If the control command of the read block is not the end command (S2: NO), the CPU 31 starts clocking (S3). The CPU 31 executes first drive condition processing in the control command of the read block (S4). In S4, the CPU 31 drives the Z-axis motor 51 based on a first drive condition of the control command.

[0079] The CPU 31 determines whether a time (4 hours) indicated by the control command has elapsed since the start of the clocking in S3 (S5). If it is determined that the time indicated by the control command has not elapsed (S5: NO), the CPU 31 determines whether a predetermined time (5 minutes) has elapsed since the start of the first drive condition processing (S6). If it is determined that the predetermined time has not elapsed since the start of the first drive condition processing (S6: NO), the CPU 31 returns the processing to S4.

[0080] If it is determined that the predetermined time has elapsed since the start of the first drive condition processing (S6: YES), the CPU 31 executes second drive condition processing (refer to FIG. 9; described later) (S8). In the second drive condition processing described later, the numerical control device 30 executes a tool change and specifies the expansion and contraction amount L.sub.b. After the second drive condition processing is ended, the CPU 31 returns the processing to S4.

[0081] Referring to FIG. 9 and FIG. 10, the second drive condition processing (S8; refer to FIG. 8) executed in the parameter estimation processing is described. A variable j that may be used in the second drive condition processing is an integer equal to or greater than 1, and indicates the number of times the tool change has been executed. The variable j is stored in the RAM 33, and has a value set to 1 after the machine tool 1 is powered on.

[0082] The CPU 31 rotates the Z-axis motor 51 in a positive direction and starts lifting of the main shaft 9 (S21). The CPU 31 determines whether the main shaft 9 has reached the ATC origin based on the rotation angle θ of the Z-axis motor 51 detected by the encoder 71 (S22).

[0083] If it is determined that the main shaft 9 has not reached the ATC origin (S22: NO), the CPU 31 returns the processing to S22. If it is determined that the main shaft 9 has reached the ATC origin (S22: YES), the CPU 31 stops the rotation of the Z-axis motor 51 and stops the lifting of the main shaft 9 (S23).

[0084] The CPU 31 rotates the Z-axis motor 51 in a reverse direction and starts lowering of the main shaft 9 (S24). The CPU 31 starts acquisition of the rotation angle θ of the Z-axis motor 51 and the disturbance force F (S25). After executing the processing of S25, the CPU 31 acquires the rotation angle θ of the Z-axis motor 51 and the disturbance force F in the period t.sub.m (=0.5 ms).

[0085] The CPU 31 determines whether the rotation angle θ of the Z-axis motor 51 and the disturbance force F have been newly acquired in the processing of S25 (S26). If it is determined that the rotation angle θ and the disturbance force F have not been newly acquired (S26: NO), the CPU 31 returns the processing to S26. If it is determined that the rotation angle θ and the disturbance force F have been newly acquired in the processing of S25 (S26: YES), the CPU 31 determines whether the main shaft 9 has reached the machining origin based on the acquired rotation angle θ (S27).

[0086] If it is determined that the main shaft 9 has not reached the machining origin (S27: NO), the CPU 31 applies low-pass filter processing to the rotation angle θ and the disturbance force F acquired in the processing of S25 (S28).

[0087] The CPU 31 performs time differentiation on the disturbance force F and calculates the differential value f (S29). The CPU 31 stores the rotation angle θ and the differential value f in association with each other in the RAM 33 (S30). The CPU 31 returns the processing to S26.

[0088] If it is determined that the main shaft 9 has reached the machining origin (S27: YES), the CPU 31 stops the acquisition of the rotation angle θ of the Z-axis motor 51 and the disturbance

force **F** started in the processing of **S25** (**S31**). The CPU **31** stops the rotation of the Z-axis motor **51** and stops the lowering of the main shaft **9** (**S32**).

[0089] The CPU **31** decides, as a peak angle $\phi_{\text{sub},j}$, the rotation angle θ corresponding to a minimum differential value f among the differential values f stored in the RAM **33** (**S33**). The CPU **31** stores the peak angle $\phi_{\text{sub},j}$ in the RAM **33** (**S34**). The CPU **31** transitions the processing to **S41** (refer to FIG. **10**).

[0090] As shown in FIG. **10**, in the processing of **S41**, based on the value of the variable j stored in the RAM **33**, the CPU **31** determines whether tool change processing being executed is the first processing after the machine tool **1** is powered on (**S41**).

[0091] If the value of the variable j is 1 and it is determined that the tool change processing being executed is the first processing after the machine tool **1** is powered on (**S41**: YES), the CPU **31** decides the peak angle $\phi_{\text{sub},j}$ decided in **S33** as the reference peak angle $\phi_{\text{sub},b}$ and stores it in the RAM **33** (**S42**). The CPU **31** adds 1 to the value of the variable j (**S46**) and returns the processing to the parameter estimation processing.

[0092] If the value of the variable j is 2 or more and it is determined that the tool change processing being executed is not the first processing after the machine tool **1** is powered on (**S41**: NO), the CPU **31** calculates the difference $\rho (= \phi_{\text{sub},j} - \phi_{\text{sub},b})$ between the peak angle $\phi_{\text{sub},j}$ decided in **S33** and the reference peak angle $\phi_{\text{sub},b}$ decided in **S42** (**S43**).

[0093] Based on the difference ρ calculated in the processing of **S43**, the CPU **31** calculates a change amount in the position of the main shaft **9** at the peak angle ϕ , and specifies this change amount as the expansion and contraction amount $L_{\text{sub},b}$ of the ball screw **41** (**S44**). The CPU **31** stores the specified expansion and contraction amount $L_{\text{sub},b}$ and an elapsed time since the start of execution of the parameter estimation processing in association with each other in the storage device **34** (**S45**). The CPU **31** adds 1 to the value of the variable j (**S46**) and returns the processing to the parameter estimation processing.

[0094] As shown in FIG. **8**, if it is determined that the time (4 hours) indicated by the control command has elapsed (**S5**: YES), the CPU **31** adds 1 to the value of the variable K (**S10**) and returns the processing to **S1**.

[0095] If the control command of the block read in the processing of **S1** is the end command (**S2**: YES), the CPU **31** estimates a parameter (coefficients P_r and F_c , thermal resistances $R_{\text{sub},\text{max}}$ and $R_{\text{sub},\text{min}}$, and boundary speed $V_{\text{sub},\text{ds}}$) (**S11**).

[0096] A method for estimating a parameter is described by way of a specific example. For example, when the processing of **S11** is started, the expansion and contraction amount $L_{\text{sub},b}$ of the ball screw **41** that varies according to time as shown in solid lines in FIG. **11** is stored in the storage device **34**. In **S11**, the CPU **31** estimates a parameter using, for example, the steepest descent method, so that the expansion and contraction amount $L_{\text{sub},b}$ (solid line in FIG. **11**) specified in **S44** (refer to FIG. **10**) approximates the expansion and contraction amount $L_{\text{sub},b}$ estimated by Expression 15. The torque $T_{\text{sub},m}$ of the Z-axis motor **51** and the feed speed $V_{\text{sub},m}$ of the main shaft **9**, which are necessary for estimating the expansion and contraction amount $L_{\text{sub},b}$ in Expression 15, are acquired in period acquisition processing (refer to FIG. **12**) described later.

[0097] The CPU **31** stores the parameter estimated in the processing of **S11** in the storage device **34** (**S12**). The CPU **31** ends the parameter estimation processing.

Period Acquisition Processing

Referring to FIG. **12**, the period acquisition processing executed by the CPU **31** of the numerical control device **30** is described. In the period acquisition processing, the CPU **31** acquires the torque $T_{\text{sub},m}$ of the Z-axis motor **51** and the feed speed $V_{\text{sub},m}$ of the main shaft **9** that are necessary for estimating the expansion and contraction amount $L_{\text{sub},b}$ in Expression 15. The torque $T_{\text{sub},m}$ and the feed speed $V_{\text{sub},m}$ are lists storing time-series data for each control period. After the machine tool **1** is powered on, the CPU **31** reads out and executes a control program stored in the

ROM 32, thereby starting the period estimation processing.

[0098] The CPU 31 starts clocking (S51). The CPU 31 determines whether the period $t_{sub.m}$ (=5 ms) has elapsed since the start of the clocking (S52). If it is determined that the period t_m has not elapsed (S52: NO), the CPU 31 returns the processing to S52.

[0099] The CPU 31 acquires the rotation angle θ , the torque $T_{sub.m}$ of the Z-axis motor 51, and the feed speed $V_{sub.m}$ of the main shaft 9 (S53). In the processing of S53, the CPU 31 acquires the rotation angle θ of the Z-axis motor 51 from the encoder 71. By acquiring the rotation angle θ and acquiring a drive current of the Z-axis motor 51 from the drive circuit 61, the CPU 31 acquires the torque $T_{sub.m}$. Based on an angular speed ($\dot{\theta}$ with one dot above) calculated from the rotation angle θ and the lead of the ball screw 41, the CPU 31 acquires the feed speed $V_{sub.m}$.

[0100] The CPU 31 stores the torque $T_{sub.m}$ and the feed speed $V_{sub.m}$ acquired in S53 in the RAM 33 (S54). The CPU 31 resets the clocking and restarts the clocking of the period t_m (S55). The CPU 31 returns the processing to S52. Every time the period t_m elapses, the CPU 31 acquires the torque $T_{sub.m}$ and the feed speed $V_{sub.m}$.

[0101] Since the parameter (coefficients P_r and F_c , thermal resistances $R_{sub.max}$ and $R_{sub.min}$, and boundary speed V_{as}) is estimated in the parameter estimation processing and the torque $T_{sub.m}$ of the Z-axis motor 51 and the feed speed $V_{sub.m}$ are acquired in the period acquisition processing, the numerical control device 30 is able to estimate the expansion and contraction amount $L_{sub.b}$ of the ball screw 41 based on Expression 15. By executing expansion and contraction amount estimation processing described later, the numerical control device 30 estimates the expansion and contraction amount $L_{sub.b}$. In the expansion and contraction amount estimation processing, the numerical control device 30 estimates the expansion and contraction amount $L_{sub.b}$ in the estimation period $t_{sub.s}$ (30 s). The estimation period $t_{sub.s}$ is set to a longer time relative to the control period (0.5 ms).

Expansion and Contraction Amount Estimation Processing

Referring to FIG. 13, the expansion and contraction amount estimation processing executed by the CPU 31 of the numerical control device 30 is described. The operator inputs an expansion and contraction amount estimation command to the input part 37. The input part 37 accepts the input expansion and contraction amount estimation command and inputs it to the CPU 31. In the case where the expansion and contraction amount estimation command is input, the CPU 31 reads out and executes a control program stored in the ROM 32, thereby starting the expansion and contraction amount estimation processing. When the expansion and contraction amount estimation processing is started, the parameter estimated in the parameter estimation processing is stored in the storage device 34.

[0102] The CPU 31 starts clocking (S61). The CPU 31 determines whether the estimation period $t_{sub.s}$ (=30 s) has elapsed since the start of the clocking (S62). If it is determined that the estimation period $t_{sub.s}$ has not elapsed (S62: NO), the CPU 31 returns the processing to S62.

[0103] If it is determined that the estimation period $t_{sub.s}$ has elapsed (S62: YES), the CPU 31 acquires, as information about the Z-axis motor 51, the torque $T_{sub.m}$ of the Z-axis motor 51 and the feed speed $V_{sub.m}$ of the main shaft 9 that are stored in the RAM 33 in S54 (refer to FIG. 12), as well as the parameter estimated in S11 (refer to FIG. 8) (S63). Based on the torque $T_{sub.m}$, feed speed $V_{sub.m}$, and parameter acquired in S63, as well as Expressions 4, 7, and 9, the CPU 31 calculates the amounts of heat $Q_{sub.m}$, $Q_{sub.f}$, and $Q_{sub.ba}$ (S64).

[0104] Based on the amounts of heat $Q_{sub.m}$, $Q_{sub.f}$, and $Q_{sub.ba}$ calculated in the processing of S64, as well as Expression 15, the CPU 31 estimates the expansion and contraction amount $L_{sub.b}$ of the ball screw 41 (S65). The CPU 31 stores the expansion and contraction amount $L_{sub.b}$ estimated in the processing of S65 in the RAM 33 (S66). The CPU 31 resets the clocking and restarts the clocking of the estimation period $t_{sub.s}$ (S67). The CPU 31 returns the processing to S62. Every time the estimation period $t_{sub.s}$ elapses, the CPU 31 estimates the expansion and contraction amount $L_{sub.b}$.

[0105] As mentioned above, the peak angle ϕ changes according to the mounted state. The numerical control device **30** determines the mounted state based on a change in peak angle ϕ for each mounting operation. In the tool change processing described later, the numerical control device **30** corrects the peak angle ϕ based on the estimated expansion and contraction amount $L_{sub.b}$ and determines the mounted state.

Tool Change Processing

Referring to FIG. **14** and FIG. **15**, the tool change processing executed by the CPU **31** of the numerical control device **30** is described. In the tool change processing, the CPU **31** performs a change of the tool **3** by the tool changer **20** and determines the mounted state in the mounting operation. The operator inputs a tool change command to the input part **37**. The input part **37** accepts the input tool change command and inputs it to the CPU **31**. In the case where the tool change command is input, the CPU **31** reads out and executes a control program stored in the ROM **32**, thereby starting the tool change processing.

[0106] In the tool change processing, the main shaft **9** is located at the machining origin. A part of the processing in the tool change processing is common with the second drive condition processing (refer to FIG. **9** and FIG. **10**). With respect to the processing executed in the tool change processing that is common with the second drive condition processing, the description will be simplified.

[0107] As shown in FIG. **14**, the CPU **31** starts lifting of the main shaft **9** (S71). The CPU **31** determines whether the main shaft **9** has reached the ATC origin (S72). If it is determined that the main shaft **9** has not reached the ATC origin (S72: NO), the CPU **31** returns the processing to S72. If it is determined that the main shaft **9** has reached the ATC origin (S72: YES), the CPU **31** stops the lifting of the main shaft **9** (S73).

[0108] The CPU **31** acquires tool information indicating the tool **3** (second tool) to be mounted on the main shaft **9** by tool change (S74). The tool information includes the number of times of mounting the tool **3**. The CPU **31** drives the magazine motor **55** to rotate the tool magazine **21** and index the grip arm **23** that grips the second tool to the tool change position (S75).

[0109] The CPU **31** starts lowering of the main shaft **9** (S76). The CPU **31** starts acquisition of the rotation angle θ of the Z-axis motor **51** and the disturbance force F (S77). In the processing of S77, by acquiring the torque $T_{sub.m}$ of the Z-axis motor **51** stored in S54 (refer to FIG. **12**) of the period acquisition processing, the CPU **31** acquires the disturbance force F included in the torque $T_{sub.m}$ based on Expression 1.

[0110] The CPU **31** determines whether the rotation angle θ of the Z-axis motor **51** and the disturbance force F have been newly acquired in the processing of S77 (S78). If it is determined that the rotation angle θ and the disturbance force F have not been newly acquired (S78: NO), the CPU **31** returns the processing to S78. If it is determined that the rotation angle θ and the disturbance force F have been newly acquired in the processing of S77 (S78: YES), the CPU **31** determines whether the main shaft **9** has reached the machining origin based on the acquired rotation angle θ (S79).

[0111] If it is determined that the main shaft **9** has not reached the machining origin (S79: NO), the CPU **31** applies low-pass filter processing to the rotation angle θ and the disturbance force F acquired in the processing of S77 (S80).

[0112] The CPU **31** performs time differentiation on the disturbance force F and calculates the differential value f (S81). The CPU **31** stores the rotation angle θ and the differential value f in association with each other in the RAM **33** (S82). The CPU **31** returns the processing to S78.

[0113] If it is determined that the main shaft **9** has reached the machining origin (S79: YES), the CPU **31** stops the acquisition of the rotation angle θ of the Z-axis motor **51** and the disturbance force F started in the processing of S77 (S83). The CPU **31** stops the lowering of the main shaft **9** (S84). The CPU **31** transitions the processing to S91 (refer to FIG. **15**).

[0114] As shown in FIG. **15**, the CPU **31** decides, as a peak angle $\phi_{sub.t}$, the rotation angle θ corresponding to the differential value f indicating the minimum among the differential values f

stored in the RAM **33** (S91). The CPU **31** acquires, among the expansion and contraction amounts L.sub.b of the ball screw **41** estimated in the expansion and contraction amount estimation processing and stored in the RAM **33**, a latest expansion and contraction amount L.sub.b1 and an expansion and contraction amount L.sub.b2 stored immediately before the expansion and contraction amount L.sub.b1 (S92). The CPU **31** calculates a difference $\Delta L.sub.b$ ($=L.sub.b1-L.sub.b2$) between the expansion and contraction amount L.sub.b1 and the expansion and contraction amount L.sub.b2 (S93).

[0115] The CPU **31** determines whether the difference $\Delta L.sub.b$ calculated in S93 is less than or equal to a threshold $\Delta L.sub.bmax$ (S94). If the difference $\Delta L.sub.b$ being a change amount in the expansion and contraction amount L.sub.b of the ball screw **41** is greater than the threshold $\Delta L.sub.bmax$, since the ball screw **41** significantly expands and contracts during a while from the tool change processing when the expansion and contraction amount L.sub.b2 is estimated to the tool change processing currently being executed, there is a risk that the position of the main shaft **9** cannot be accurately controlled in the machining of the workpiece performed during this while. If it is determined that the difference $\Delta L.sub.b$ is greater than the threshold $\Delta L.sub.bmax$ (S94: NO), the CPU **31** executes notification processing (S103). In the processing of S103, the CPU **31** performs, as the notification processing, an error notification through the display part **38**. The CPU **31** ends the tool change processing.

[0116] If it is determined that the difference $\Delta L.sub.b$ is less than or equal to the threshold $\Delta L.sub.bmax$ (S94: YES), the CPU **31** corrects the peak angle $\phi.sub.t$ decided in the processing of S91 based on the expansion and contraction amount L.sub.b1 acquired in the processing of S92, and decides a corrected peak angle $\phi.sub.d$ (S95). In the processing of S95, by inverting the calculation of the expansion and contraction amount L.sub.b from the difference ρ performed in the processing of S44, the CPU **31** calculates the difference ρ of the peak angle ϕ from the expansion and contraction amount L.sub.b1 of the ball screw **41**. The CPU **31** decides the corrected peak angle $\phi.sub.d$ by subtracting the calculated difference ρ from the peak angle $\phi.sub.t$. The CPU **31** stores the corrected peak angle $\phi.sub.d$ decided in the processing of S95 and the number of times of mounting the tool **3** acquired in the processing of S74 in association with each other in the storage device **34** (S96).

[0117] The CPU **31** determines whether there is a tool **3** at the tool change position when the main shaft **9** is lowered (S97). If it is determined that there is no tool **3** at the tool change position (S97: NO), the CPU **31** executes the notification processing (S103) and ends the tool change processing. If it is determined that there is a tool **3** at the tool change position (S97: YES), the CPU **31** transitions the processing to S98.

[0118] The CPU **31** determines whether the number of target peak angles (peak angles among the stored corrected peak angles $\phi.sub.d$, excluding a first peak angle $\phi.sub.d1$ and the peak angle $\phi.sub.d$ in the case where the incompletely mounted state is reached) among the corrected peak angles $\phi.sub.d$ stored in the storage device **34** in the processing of S96 is equal to or greater than a predetermined number (S98). If it is determined that the number of target peak angles is less than the predetermined number (S98: NO), the CPU **31** ends the tool change processing.

[0119] If it is determined that the number of target peak angles is equal to or greater than the predetermined number (S98: YES), the CPU **31** executes statistical processing (S99). In the processing of S99, based on the number of times of mounting, the CPU **31** acquires the predetermined number of target peak angles in a descending order of newness from a second peak angle among the target peak angles stored in the storage device **34**. The CPU **31** calculates the average value μ and standard deviation σ of the acquired target peak angles. Based on the average value μ and standard deviation σ calculated in the processing of S99, the CPU **31** decides $\mu+6\sigma$ as the determination threshold $\phi.sub.max$ (S100).

[0120] The CPU **31** determines whether the first peak angle $\phi.sub.d1$ among the peak angles $\phi.sub.d$ stored in the storage device **34** in the processing of S96 is less than or equal to the

determination threshold $\phi_{\text{sub.max}}$ decided in the processing of S100 (S101). If it is determined that the first peak angle $\phi_{\text{sub.d1}}$ is greater than the determination threshold $\phi_{\text{sub.max}}$ (S101: NO), the CPU 31 determines that the mounted state after the mounting operation is an incompletely mounted state. In this case, the CPU 31 deletes the first peak angle $\phi_{\text{sub.d1}}$ stored in the storage device 34 (S102). The CPU 31 executes the notification processing (S103) and ends the tool change processing.

[0121] If it is determined that the first peak angle $\phi_{\text{sub.d1}}$ is less than or equal to the determination threshold $\phi_{\text{sub.max}}$ (S101: YES), the CPU 31 determines that the mounted state after the mounting operation is a completely mounted state. In this case, the CPU 31 ends the tool change processing.

Actions and Effects of Present Embodiment

As described above, the temperature T_b of the ball screw 41 changes due to heating from the outside and heat dissipation to the outside. When the temperature change ΔT_b occurs, the ball screw 41 expands and contracts in the Z-axis direction. When the ball screw 41 expands and contracts, the peak angle ϕ changes according to the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41. In the determination of the mounted state based on the peak angle ϕ , it is difficult to distinguish between a variation in the peak angle ϕ due to the expansion and contraction of the ball screw 41 and a variation in the peak angle ϕ due to the incompletely mounted state.

[0122] The CPU 31 of the numerical control device 30 estimates the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41 (S65). The CPU 31 acquires the disturbance force F of the Z-axis motor 51 (S77). In the processing of S77, by acquiring the torque $T_{\text{sub.m}}$ of the Z-axis motor 51, the CPU 31 acquires the disturbance force F included in the torque $T_{\text{sub.m}}$. Based on the peak angle ϕ , which is decided by the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41 estimated in the processing of S65 and a variation in the differential value f being a time differential of the disturbance force F , the CPU 31 determines the mounted state (S101). In this way, the numerical control device 30 estimates the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41. Based on the estimated expansion and contraction amount $L_{\text{sub.b}}$ and the torque $T_{\text{sub.m}}$ of the Z-axis motor 51, the numerical control device 30 determines the mounted state of the tool 3 with respect to the main shaft 9. Since the numerical control device 30 takes into consideration the expansion and contraction amount $L_{\text{sub.b}}$ in the determination of the mounted state, the mounted state can be accurately determined.

[0123] The CPU 31 acquires the information (torque $T_{\text{sub.m}}$ of Z-axis motor 51 and feed speed $V_{\text{sub.m}}$ of main shaft 9) about the Z-axis motor 51 (S63). Based on the acquired torque $T_{\text{sub.m}}$ and feed speed $V_{\text{sub.m}}$, the CPU 31 calculates the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$ (S64). Based on the calculated amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$, as well as Expression 15, the CPU 31 estimates the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41 (S65). The numerical control device 30 acquires the information about the Z-axis motor 51 resulting from heat that enters and exits the ball screw 41, and, by calculating the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$ based on the acquired information, estimates the expansion and contraction amount $L_{\text{sub.b}}$ of the ball screw 41. Thus, the numerical control device 30 is able to accurately estimate the expansion and contraction amount $L_{\text{sub.b}}$.

[0124] The CPU 31 acquires, as the information about the Z-axis motor 51, the torque $T_{\text{sub.m}}$ of the Z-axis motor 51 and the feed speed $V_{\text{sub.m}}$ of the main shaft 9 (S63). By acquiring the angular speed (θ with one dot above), the CPU 31 acquires the feed speed $V_{\text{sub.m}}$. Based on the acquired torque $T_{\text{sub.m}}$ and feed speed $V_{\text{sub.m}}$, the numerical control device 30 is able to accurately calculate the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$ that enter and exit the ball screw 41.

[0125] The CPU 31 acquires the rotation angle θ of the Z-axis motor 51 and the disturbance force F (S77). The CPU 31 performs time differentiation on the disturbance force F and calculates the differential value f (S80). The CPU 31 decides, as the peak angle $\phi_{\text{sub.t}}$, the rotation angle θ corresponding to the differential value f indicating the minimum among the calculated differential

values f (S91). The CPU 31 corrects the decided peak angle ϕ , based on the expansion and contraction amount $L_{sub.b1}$, and decides the corrected peak angle $\phi_{sub.d}$ (S95). By determining whether the first peak angle $\phi_{sub.d1}$ among the corrected peak angles $\phi_{sub.d}$ is less than or equal to the determination threshold $\phi_{sub.max}$, the CPU 31 determines the mounted state (S101). According to the corrected peak angle $\phi_{sub.d}$ obtained by correcting the peak angle ϕ based on the expansion and contraction amount $L_{sub.b1}$ of the ball screw 41, the numerical control device 30 determines the mounted state. Accordingly, since a deviation between the position of the main shaft 9 estimated from the peak angle $\phi_{sub.d}$ and the actual position of the main shaft 9 is suppressed, the numerical control device 30 is able to accurately determine the mounted state of the tool 3 with respect to the main shaft 9.

[0126] The CPU 31 stores the corrected peak angle $\phi_{sub.d}$ and the number of times of mounting the tool 3 in association with each other in the storage device 34 (S96). The CPU 31 acquires a predetermined number of target peak angles stored in the storage device 34 and calculates the average value μ and standard deviation σ of the acquired target peak angles (S99). By determining whether the first peak angle $\phi_{sub.d1}$ is less than or equal to the determination threshold $\phi_{sub.max}$ ($=\mu+6\sigma$), the CPU 31 determines the mounted state (S101). The numerical control device 30 determines the mounted state based on the statistical values (average value μ and standard deviation σ) on the basis of the predetermined number of target peak angles among the corrected peak angles $\phi_{sub.d}$ stored in the storage device 34. Accordingly, the numerical control device 30 is able to eliminate the influence of variations in determining the corrected peak angle $\phi_{sub.d}$ and accurately determine the mounted state.

[0127] The CPU 31 determines whether the difference $\Delta L_{sub.b}$ between the expansion and contraction amount $L_{sub.b1}$ and the expansion and contraction amount $L_{sub.b2}$ is less than or equal to the threshold $\Delta L_{sub.bmax}$ (S94). If it is determined that the difference $\Delta L_{sub.b}$ is less than or equal to the threshold $\Delta L_{sub.bmax}$ (S94: YES), by determining whether the first peak angle $\phi_{sub.d1}$ is less than or equal to the determination threshold $\phi_{sub.max}$, the CPU 31 determines the mounted state (S101). If it is determined that the difference $\Delta L_{sub.b}$ is greater than the threshold $\Delta L_{sub.bmax}$ (S94: NO), the CPU 31 ends the tool change processing without determining the mounted state. If the difference $\Delta L_{sub.b}$ being the change amount in the expansion and contraction amount $L_{sub.b}$ of the ball screw 41 is greater than the threshold $\Delta L_{sub.bmax}$, there is a risk that the expansion and contraction amount $L_{sub.b}$ may not be accurately estimated. In the numerical control device 30, by determining the magnitude of the difference $\Delta L_{sub.b}$ of the expansion and contraction amount $L_{sub.b}$, erroneous determination of the mounted state due to inaccurate estimation of the expansion and contraction amount $L_{sub.b}$ can be suppressed.

[0128] In the expansion and contraction amount estimation processing, the CPU 31 estimates the expansion and contraction amount $L_{sub.b}$ every estimation period $t_{sub.s}$. Since the estimation period $t_{sub.s}$ ($=30$ s) is sufficiently long relative to the control period ($=0.5$ ms), the numerical control device 30 is able to accurately estimate the expansion and contraction amount $L_{sub.b}$ with a small amount of calculation.

[0129] The CPU 31 repeatedly acquires the torque $T_{sub.m}$ of the Z-axis motor 51 as well as the feed speed $V_{sub.m}$ and rotation angle θ of the main shaft 9 every period $t_{sub.m}$ (S53). The period $t_{sub.m}$ is a shortest time among the control periods in the numerical control device 30. The numerical control device 30 acquires the torque $T_{sub.m}$ every control period, the control period being a smallest unit at which the torque $T_{sub.m}$ can be acquired. The numerical control device 30 decides the peak angle ϕ based on the disturbance force F included in the acquired torque $T_{sub.m}$, and determines the mounted state. Since the torque $T_{sub.m}$ is acquired at intervals of the control period, the numerical control device 30 is able to relatively correctly decide the peak angle ϕ and relatively accurately determine the mounted state.

Modifications

The present disclosure can be modified in various ways from the above-described embodiment. The various modifications described hereinafter can be combined with each other as long as no contradictions arise. For example, the numerical control device **30** is not limited to being provided in the machine tool **1**, and may be provided separately from the machine tool **1**. For example, the numerical control device **30** may be a device (such as a personal computer (PC) or a special purpose machine) connected to the machine tool **1**. The tool changer **20** is not limited to being of the turret type, and may be of, for example, an arm type.

[0130] The CPU **31** of the numerical control device **30** may acquire the torque $T_{\text{sub.m}}$ of the Z-axis motor **51** stored in the RAM **33** in the processing of **S54**, instead of the disturbance force F , and decide the peak angle ϕ based on a variation in the torque $T_{\text{sub.m}}$. In this case, the CPU **31** performs time differentiation on the torque $T_{\text{sub.m}}$, similarly to the case of acquiring the disturbance force F . The CPU **31** decides, as the peak angle ϕ , the rotation angle θ corresponding a differential value of the torque $T_{\text{sub.m}}$ indicating the minimum among the differential values of the torque $T_{\text{sub.m}}$. The CPU **31** corrects the decided peak angle ϕ based on the expansion and contraction amount $L_{\text{sub.b}}$, and determines the mounted state. In this way, based on a variation in the torque $T_{\text{sub.m}}$, the numerical control device **30** is able to determine the mounted state.

[0131] The CPU **31** of the numerical control device **30** may determine, as a mounted state of the tool **3**, a mounted state other than the incompletely mounted state in which the tool **3** is mounted with foreign matter adhering between the mounting part **15** and the mounting hole **92**. For example, the CPU **31** may determine a tool deterioration state as the mounted state. The tool deterioration state is, for example, a state in which a connected portion of the pull stud **17** with the mounting part **15** is recessed inward due to repeated mounting of the tool **3**. In the tool deterioration state, the position where the gripper **85** pulls up and grips the pull stud **17** in the mounting operation is lower than that in the completely mounted state. Accordingly, in the case where the mounted state is the tool deterioration state, the peak angle ϕ in the case where the differential value f being a time differential of the disturbance force F indicates the minimum varies downward compared to the case where the mounted state is the completely mounted state. For example, by determining whether the first peak angle $\phi_{\text{sub.d1}}$ is equal to or greater than the threshold ($=\mu-6\sigma$) decided in the processing of **S100** with $\mu-6\sigma$ as the determination threshold, the CPU **31** may determine whether the mounted state is either the completely mounted state or the tool deterioration state.

[0132] The CPU **31** of the numerical control device **30** may estimate the expansion and contraction amount $L_{\text{sub.b}}$ without calculating the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$. For example, the numerical control device **30** may detect a temperature change of the ball screw **41** using a temperature sensor, and estimate the expansion and contraction amount $L_{\text{sub.b}}$ based on the temperature change. For example, the numerical control device **30** may detect a change in a predetermined position at an upper end of the ball screw **41** using a displacement sensor, and estimate the expansion and contraction amount $L_{\text{sub.b}}$ based on the change in the predetermined position at the upper end.

[0133] The mathematical model for estimating the expansion and contraction amount $L_{\text{sub.b}}$ is an example and may be modified as appropriate. The numerical control device **30** does not have to calculate all of the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$. For example, the numerical control device **30** may estimate the expansion and contraction amount $L_{\text{sub.b}}$ based on the calculated amounts of heat $Q_{\text{sub.m}}$ and $Q_{\text{sub.ba}}$ without calculating the amount of heat $Q_{\text{sub.f}}$. The numerical control device **30** may calculate the amounts of heat $Q_{\text{sub.m}}$, $Q_{\text{sub.f}}$, and $Q_{\text{sub.ba}}$ for each of multiple sections obtained by dividing the ball screw **41** in the Z-axis direction, and estimate the expansion and contraction amount $L_{\text{sub.b}}$.

[0134] In the processing of **S63**, the CPU **31** of the numerical control device **30** may acquire information about the Z-axis motor **51** different from the torque $T_{\text{sub.m}}$ of the Z-axis motor **51** and the feed speed $V_{\text{sub.m}}$. For example, in the processing of **S63**, the CPU **31** may acquire the temperature of the Z-axis motor **51** from a sensor that detects the temperature of the Z-axis motor

51.

[0135] The CPU **31** of the numerical control device **30** may generate an interpolation angle and an interpolation disturbance force obtained by interpolating (oversampling) the rotation angle θ and the disturbance force F and increase resolution. In this case, by deciding the peak angle ϕ based on a differential value obtained by time differentiation of the interpolation disturbance force and the interpolation angle, the CPU **31** is able to approximate the peak angle ϕ relatively closely to the rotation angle θ at the timing at which the plate cam **47** separates from the cam follower **49**.

[0136] The CPU **31** of the numerical control device **30** may decide, as the peak angle ϕ , the rotation angle θ in the case where the differential value f indicates the maximum in the mounting operation. The CPU **31** may determine the mounted state based on the position of the main shaft **9** in the Z-axis direction estimated based on the rotation angle θ of the Z-axis motor **51**. In this case, the CPU **31** corrects a peak position corresponding to the differential value f indicating the minimum among the differential values f by the estimated expansion and contraction amount $L_{\text{sub.b}}$, and determines the mounted state.

[0137] It is sufficient if the correction of the peak angle $\phi_{\text{sub.t}}$ is based on the expansion and contraction amount $L_{\text{sub.b}}$, and a method for deciding the corrected peak angle $\phi_{\text{sub.d}}$ is not limited to the above-described embodiment. For example, the CPU **31** may extract a predetermined number of expansion and contraction amounts $L_{\text{sub.b}}$ from the expansion and contraction amounts $L_{\text{sub.b}}$ stored in the RAM **33**, calculate the difference ρ of the peak angle ϕ from an average of the extracted expansion and contraction amounts $L_{\text{sub.b}}$, and subtract the difference ρ from the peak angle $\phi_{\text{sub.t}}$, thereby deciding the corrected peak angle $\phi_{\text{sub.d}}$. In the above-described embodiment, the difference ρ of the peak angle ϕ is calculated from the expansion and contraction amount $L_{\text{sub.b}}$, and the peak angle $\phi_{\text{sub.t}}$ is corrected based on the difference ρ . In contrast, the CPU **31** may estimate the position of the main shaft **9** in the Z-axis direction from the peak angle $\phi_{\text{sub.t}}$, correct the estimated position of the main shaft **9** in the Z-axis direction by the expansion and contraction amount $L_{\text{sub.b}}$, and decide the corrected peak angle $\phi_{\text{sub.d}}$ from the corrected position of the main shaft **9** in the Z-axis direction. The CPU **31** may store the peak angle $\phi_{\text{sub.t}}$ for each tool **3** in the storage device **34**, and subtract the expansion and contraction amount $L_{\text{sub.b1}}$ from an average of the peak angles $\phi_{\text{sub.t}}$ for each tool **3**, thereby deciding the corrected peak angle $\phi_{\text{sub.d}}$ for each tool **3**.

[0138] In the above-described embodiment, the CPU **31** of the numerical control device **30** stores the corrected peak angle $\phi_{\text{sub.d}}$ and the number of times of mounting the tool **3** in association with each other in the storage device **34**. In contrast, as an example, the CPU **31** may store the corrected peak angle $\phi_{\text{sub.d}}$ and a use time of the tool **3** in association with each other in the storage device **34**.

[0139] The numerical control device **30** may extract a predetermined number of target peak angles in a descending order of newness from the second peak angle among the target peak angles, and calculate, as a statistical value, a value uniquely decided from the extracted predetermined number of peak angles $\phi_{\text{sub.d}}$, instead of the average value μ . For example, the numerical control device **30** may calculate, as a statistical value, the median of the extracted predetermined number of peak angles $\phi_{\text{sub.d}}$.

[0140] The magnitude of the determination threshold $\phi_{\text{sub.max}}$ is not limited to $\mu + 6\sigma$, and may be modified as appropriate. The determination threshold ϕ_{max} may be a predetermined value stored in the storage device **34**, rather than a value decided by statistical processing of a predetermined number of target peak angles.

[0141] The difference $\Delta L_{\text{sub.b}}$ calculated as the change amount in the expansion and contraction amount $L_{\text{sub.b}}$ may be a value other than $L_{\text{sub.b1}} - L_{\text{sub.b2}}$. For example, the difference $\Delta L_{\text{sub.b}}$ may be a difference between the expansion and contraction amount $L_{\text{sub.b1}}$ that is latest and an expansion and contraction amount $L1$ that is stored earliest among the expansion and contraction amounts $L_{\text{sub.b}}$ of the ball screw **41** stored in the RAM **33**.

[0142] In the expansion and contraction amount estimation processing, the magnitude of the estimation period $t_{sub.s}$ for estimating the expansion and contraction amount $L_{sub.b}$ may be modified as appropriate. The estimation period $t_{sub.s}$ may be, for example, the control period (0.5 ms).

[0143] The magnitude of the period $t_{sub.m}$ for acquiring the torque $T_{sub.m}$ of the Z-axis motor **51** and the feed speed $V_{sub.m}$ of the main shaft **9** may be modified as appropriate. The period $t_{sub.m}$ may be larger than the control period (0.5 ms).

[0144] In the above-described embodiment, in the numerical control device **30**, a method for acquiring the disturbance force F may be modified as appropriate. For example, instead of using Expression 1, the numerical control device **30** may acquire, as the disturbance force F , a difference between the torque $T_{sub.m}$ and inertial force.

[0145] In the numerical control device **30**, low-pass filter processing does not have to be applied to the rotation angle θ and the disturbance force F . In this case, the processing in **S28** and **S80** may be omitted. In the numerical control device **30**, high-pass filter or band-pass filter processing may be applied to the rotation angle θ and the disturbance force F .

[0146] In the numerical control device **30**, in the parameter estimation processing, the expansion and contraction amount $L_{sub.b}$ may be specified using a disturbance force F other than that occurring during a tool change of the tool **3**. For example, the expansion and contraction amount $L_{sub.b}$ may be specified using a disturbance force F generated by rotating the X-axis motor **53**, moving the table **13** to an opposite side of the motor and pressing it against a stroke end.

[0147] A configuration of the Z-axis movement mechanism **8** may be modified as appropriate. The Z-axis movement mechanism **8** may have a configuration in which the main shaft **9** is moved by rotation of the Z-axis motor **51** instead of the ball screw **41**. The Z-axis movement mechanism **8** may include, for example, a sliding screw extending in the Z-axis direction and a spline shaft with teeth formed in the Z-axis direction, instead of the ball screw **41**. In this case, the sliding screw and the spline shaft are an example of the “feed shaft” of the present disclosure.

[0148] In the above-described embodiment, the peak angle ϕ is decided with the differential value f of the disturbance force F included in the torque $T_{sub.m}$ as a feature value. However, the feature value for determining the peak angle ϕ may be modified as appropriate. For example, the numerical control device **30** may, without differentiating the disturbance force F , decide the peak angle ϕ with the disturbance force F (torque $T_{sub.m}$) that varies with time as a feature value. In this case, the numerical control device **30** may decide, as the peak angle ϕ , the rotation angle θ corresponding to the disturbance force F indicating either the maximum or the minimum of the disturbance force F .

[0149] The numerical control device **30** may decide the peak angle ϕ with the torque $T_{sub.m}$ that varies with time as a feature value. In this case, the numerical control device **30** may decide, as the peak angle ϕ , the rotation angle θ corresponding to the torque $T_{sub.m}$ indicating either the maximum or the minimum of the torque $T_{sub.m}$.

[0150] The numerical control device **30** may decide the peak angle ϕ with the following as a feature value: a squared value of the disturbance force F (torque $T_{sub.m}$), a value obtained by second-order differentiation of the disturbance force F (torque $T_{sub.m}$), a value obtained by averaging the disturbance force F (torque $T_{sub.m}$) within a predetermined time period, or the like.

[0151] The numerical control device **30** does not have to use the rotation angle θ corresponding to the differential value f indicating the minimum as the peak angle ϕ in the mounting operation. The numerical control device **30** may use the rotation angle θ corresponding to the differential value f indicating the maximum as the peak angle ϕ in the mounting operation. The numerical control device **30** may use the rotation angle θ corresponding to the differential value f indicating the maximum as the peak angle ϕ in the detachment operation. The numerical control device **30** may use the rotation angle θ corresponding to the differential value f indicating the minimum as the peak angle ϕ in the detachment operation.

[0152] The numerical control device **30** does not have to decide, as the peak angle ϕ , the rotation

angle θ corresponding to the differential value f indicating either the maximum or the minimum among the differential values f . For example, the numerical control device **30** may decide, as the peak angle ϕ , a rotation angle $(\theta - \theta_1)$ offset by a predetermined angle θ_1 from the rotation angle θ corresponding to the differential value f indicating the minimum. For example, the numerical control device **30** may decide, as the peak angle ϕ , a rotation angle $(\beta\theta)$ obtained by multiplying the rotation angle θ corresponding to the differential value f indicating the minimum by a predetermined coefficient β .

Others

In the above-described embodiment, the Z-axis direction is an example of the “predetermined direction” of the present disclosure. The ball screw **41** is an example of the “feed shaft” of the present disclosure. The Z-axis motor **51** is an example of the “motor” of the present disclosure. The CPU **31** executing the expansion and contraction amount estimation processing is an example of the “estimation part” of the present disclosure. The CPU **31** executing the processing of **S77** is an example of the “first acquisition part” of the present disclosure. The CPU **31** executing the processing of **S101** is an example of the “determination part”, “correction determination part”, and “statistical determination part” of the present disclosure. The CPU **31** executing the processing of **S63** is an example of the “second acquisition part” of the present disclosure. The CPU **31** executing the processing of **S64** is an example of the “calculation part” of the present disclosure. The angular speed is an example of the “rotation speed” of the present disclosure. The CPU **31** executing the processing of **S81** is an example of the “differentiation part” of the present disclosure. The CPU **31** executing the processing of **S91** is an example of the “decision part” of the present disclosure. The peak angle $\phi_{\text{sub.d}}$ is an example of the “corrected peak angle” of the present disclosure. The CPU **31** executing the processing of **S95** is an example of the “second decision part” of the present disclosure. The CPU **31** executing the processing of **S96** is an example of the “storage part” of the present disclosure. The CPU **31** executing the processing of **S99** is an example of the “statistical processing part” of the present disclosure. The difference $\Delta L_{\text{sub.b}}$ is an example of the “difference” of the present disclosure. The threshold $\Delta L_{\text{sub.max}}$ is an example of the “threshold” of the present disclosure. The period $t_{\text{sub.m}}$ is an example of the “control period” of the present disclosure. The expansion and contraction amount estimation processing is an example of the “estimation process” of the present disclosure. The processing of **S77** is an example of the “first acquisition process” of the present disclosure. The processing of **S101** is an example of the “determination process” of the present disclosure.

Claims

1. A numerical control device that controls an operation of a machine tool, the machine tool comprising a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft and allow the tool gripped by a tool changer to be mounted on the main shaft, wherein the numerical control device comprises: an estimation part, estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition part, acquiring a torque of the motor in a mounting operation of the tool; and a determination part, determining a mounted state of the tool with respect to the main shaft based on the expansion and contraction amount estimated by the estimation part and the torque acquired by the first acquisition part.
2. The numerical control device as claimed in claim 1, wherein the estimation part comprises: a second acquisition part, acquiring information about the motor that drives when moving the main shaft in the predetermined direction via the feed shaft; and a calculation part, calculating amount of heat entering and exiting the feed shaft based on the information acquired by the second acquisition part, and the estimation part estimates the expansion and contraction amount based on the amount

of heat calculated by the calculation part.

3. The numerical control device as claimed in claim 2, wherein the second acquisition part acquires, as the information, the torque and a rotation speed of the motor.

4. The numerical control device as claimed in claim 1, wherein the first acquisition part further acquires a rotation angle ϕ of the motor; and the determination part comprises: a differentiation part, differentiating the torque acquired by the first acquisition part and calculating a differential value of the torque corresponding to the rotation angle; a first decision part, deciding, as a peak angle, the rotation angle corresponding to the differential value indicating either maximum or minimum among the differential values calculated by the differentiation part; a second decision part, deciding a corrected peak angle by correcting the peak angle decided by the first decision part based on the expansion and contraction amount estimated by the estimation part; and a correction determination part, determining the mounted state based on the corrected peak angle decided by the second decision part.

5. The numerical control device as claimed in claim 4, wherein the determination part comprises: a storage part, storing the corrected peak angle decided by the second decision part and number of uses or a use time of the tool with respect to the main shaft in association with each other; a statistical processing part, performing statistical processing based on a plurality of the corrected peak angles stored by the storage part and calculating a statistical value; and a statistical determination part, determining the mounted state based on the statistical value calculated by the statistical processing part.

6. The numerical control device as claimed in claim 1, wherein, in a case where a difference between the expansion and contraction amount estimated by the estimation part for a n -th time and the expansion and contraction amount estimated by the estimation part for a $(n-1)$ -th time is less than or equal to a threshold, the determination part determines the mounted state, in which n is an integer equal to or greater than 2; in a case where the difference is greater than the threshold, the determination part does not determine the mounted state.

7. The numerical control device as claimed in claim 1, wherein the estimation part estimates the expansion and contraction amount in an estimation period longer than a control period of the numerical control device.

8. The numerical control device as claimed in claim 1, wherein the first acquisition part acquires the torque and a rotation angle for each control period of the numerical control device.

9. A determination method for a mounted state of a tool with respect to a main shaft in a machine tool, the machine tool comprising a main shaft that mounts a tool, a feed shaft extending in a predetermined direction, and a motor that rotates the feed shaft around an axis extending in the predetermined direction, in which the motor rotates the feed shaft to thereby move the main shaft to allow the tool gripped by a tool changer to be mounted on the main shaft, wherein the determination method comprises: an estimation process for estimating an expansion and contraction amount of the feed shaft in the predetermined direction; a first acquisition process for acquiring a torque of the motor in a mounting operation of the tool; and a determination process for determining the mounted state based on the expansion and contraction amount estimated by the estimation process and the torque acquired by the first acquisition process.
